

## Judgment about Design Viability (F1)

"Design viability" is the capacity of the proposed solution to solve the problem. Design viability refers to how your design decisions are tethered to your design requirements. For example, if one of your design requirements is that the total cost of the prototype cannot exceed X, then you could present a table of costs for each component of your proposed design solution as justification for why your design is likely to meet that requirement.

"Credible evidence" means it shouldn't just be your hunch that this will work. Be creative about how you validate the viability of your design. You should go beyond just having a stakeholder or expert reviewer say that the design seems viable. For example, you could point to evidence that a sub-system already functions, or you demo a proof of concept, or you've discussed with an expert that said, "I've seen something similar work in this other related situation...", etc.

Use as many rows as needed. Remember, this chart below may help you organize your ideas, but you will need to actually execute the "credible evidence."

| Design Viability Description                | Connection to Design Requirement                                                                             | Credible Evidence?                                                          |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| <i>Example:</i> The solution is inexpensive | <i>Example:</i> Design does not cost more than \$X                                                           | <i>Example:</i> Create a table of costs for each component of the solution. |
| Our solution is inexpensive                 | Our design cost was under \$100                                                                              | Wood for stability-FREE                                                     |
| Our solution works when needed.             | Our design had the right equipment to work                                                                   | Cover-FREE                                                                  |
| Our solution keeps the flower healthy       | Our design had cover for the flower to be as healthy as possible.                                            | Flower-\$5                                                                  |
| Our solution was one of the best projects.  | Our design was the last one done since we spent too much time planning and making adjustments on the project | Battery-\$20                                                                |